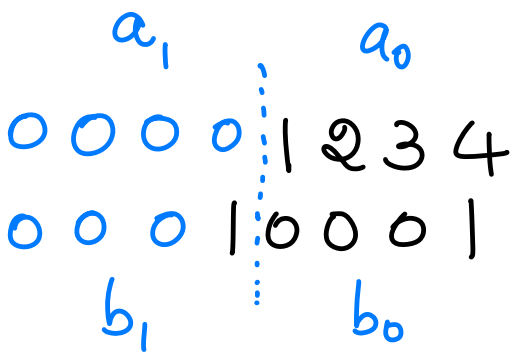
Karatsuba's Algorithm

(Multiplication of Large Numbers)



$$a = a_1 a_0 = a_1 \times 10^4 + a_0$$

$$b = b_1 b_0 = b_1 \times 10^4 + b_0$$

$$a*b = (a_1 10^4 + a_0)(b_1 10^4 + b_0)$$

$$= \underbrace{a_1 b_1}_{c_2} 10^8 + \underbrace{(a_0 b_1 + a_1 b_0)}_{c_1} 10^4 + \underbrace{a_0 b_0}_{c_0}$$

c_2: $a_1 \times b_1 = 0000 \times 0001 = 0$

c_0: $a_0 \times b_0 = 1234 \times 0001 = 1234$

$$c_1 = (a_1 + a_0) * (b_1 + b_0) - c_0 - c_2$$

$$= (0000 + 1234) * (0001 + 0001) - 1234 + 0$$

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$$= (1234) * (0002) - 1234 + 0$$

$$= 2468 - 1234$$

$$= 1234$$

$$a * b = 0 \times 10^{8} + (1234) \times 10^{4} + 1234$$

$$= 12340000 + 1234$$

$$= 12341234.$$

Brute Force (usual) Multiplication | takes $O(n^2)$ for n-digit multiplication.

Karatsuba's Algorithm

Recurrence for usual multiplication

$$T(n) = 4T(n/2)$$

$$= 2^2\left(4T(n/4)\right)$$

$$= 2^4\left(4T(n/8)\right)$$

$$= 2^6\left(4T(n/16)\right)$$

$$= 2^8 T\left(n/2^4\right)$$

$$\vdots$$

$$= 2^9 T\left(n/2^k\right)$$

$$\vdots$$

$$1.$$

$$2^k = n$$

$$\log(2^k) = \log(n)$$

$$k = \log_2 n.$$

$$T(1) = 1.$$

$$\simeq 2^{2(\log_2 n)} \cdot 1$$

$$= 2^{\log_2 n^2}$$

$$2^{\log_2 n} = n$$

$$\simeq n^2 \in \Theta(n^2)$$

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Karatsuba's Algorithm:

$$T(n) = 3T\left(n/2\right)$$

$$= 3\left(3T\left(n/4\right)\right)$$

$$= 3^2 T\left(n/2^2\right)$$

$$= 3^2\left(3T\left(n/8\right)\right)$$

$$= 3^3 T\left(n/2^3\right)$$

$$\vdots$$

$$= 3^k T\left(n/2^k\right)$$

$2^k = n$

$k = \log_2 n$

$$= 3^{\log_2 n} \cdot 1$$

$$= 3^{\log_2 n} = n^x$$

$$3^{\log_2 n} = n^x$$

$$\log_3\left(3^{\log_2 n}\right) = \log_3\left(n^x\right)$$

$$\log_2 n \left(\log_3 3\right) = x\left(\log_3 n\right)$$

$$\log_2 n = x \log_3 n$$

$$\Rightarrow \quad x = \frac{\log_2 n}{\log_3 n}$$

$$x = \frac{\frac{\log_e n}{\log_e 2}}{\frac{\log_e n^3}{\log_e 3}}$$

$$x = \frac{\log_e 3}{\log_e 2} = \log_2 3$$

$$= 1.58\ldots$$

$$\therefore T(n) \in \Theta\left(n^{\log_2 3}\right) = \Theta\left(n^{1.58\ldots}\right)$$

Strassen's multiplication:

$$\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix}\begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix} = \begin{bmatrix} a_{00}b_{00}+a_{01}b_{10} & a_{00}b_{01}+a_{01}b_{11} \\ a_{10}b_{00}+a_{11}b_{10} & a_{10}b_{01}+a_{11}b_{11} \end{bmatrix}$$

2 2 2 2

$2 \times 4 = 8$ multiplications are used.

$$T(n) = 8T(n/2)$$

$$= 8(8(T(n/4))$$

$$= 8^2 T(n/2^2)$$

$$= 8^3 T(n/2^3)$$

$$\vdots$$

$$= 8^k T(n/2^k)$$

$$= 1$$

$$= 8^{\log_2 n} \cdot 1$$

$$= \left(2^{\log_2 n}\right)^3$$

$$= \underline{n^3} \in \Theta(n^3).$$

$2^k = n$

$k = \log_2 n$

↑ usual (brute-force)

matrix multiplication time Complexity is $\Theta(n^3)$.

Strassen's multiplication:

$$\begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix}\begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix} = \begin{bmatrix} m_1+m_4-m_5+m_7 & m_3+m_5 \\ m_2+m_4 & m_1+m_3-m_2+m_6 \end{bmatrix}$$

(i) (ii) (iii) (iv)

where

$m_1 = (a_{00}+a_{11})*(b_{00}+b_{11})$

$m_2 = (a_{10}+a_{11})*b_{00}$

$m_3 = a_{00}*(b_{01}-b_{11})$

$m_4 = a_{11}*(b_{10}-b_{00})$

$m_5 = (a_{00}+a_{01})*b_{11}$

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$$m_6 = (a_{10} - a_{00}) * (b_{00} + b_{01})$$

$$m_7 = (a_{01} - a_{11}) * (b_{10} + b_{11})$$

$$m_3 + m_5 = a_{00}b_{01} - a_{00}b_{11} + a_{00}b_{11} + a_{01}b_{11}$$

$$= a_{00}b_{01} + a_{01}b_{11}. \quad \text{- - - - (ii)}$$

H.W: Find (i) $m_1 + m_4 - m_5 + m_7$

(ii) $m_3 + m_5$ (done)

(iii) $m_2 + m_4$

(iv) $m_1 + m_3 - m_2 + m_6$

(This will prove that Strassen's matrix multiplication algorithm is correct).

Strassen's multiplications uses

7 multiplications

$$T(n) = 7T(n/2)$$

$$= 7^2 T(n/2^2)$$

$$= 7^3 T(n/2^3)$$

$$\vdots$$

$$= 7^k T(n/2^k)$$

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$$= 7^{\log_2 n} = n^x$$

$2^k = n$

$k = \log_2 n$

$$\Rightarrow \log_7\left(7^{\log_2 n}\right) = \log_7\left(n^x\right)$$

$$\Rightarrow \log_2 n = x \log_7 n.$$

$$\Rightarrow x = \frac{\log_2 n}{\log_7 n}$$

$$x = \frac{\frac{\log_e 9}{\log_e 2}}{\frac{\log_e 7}{\log_e 7}}$$

$$x = \frac{\log_e 9}{\log_e 2}$$

$$x = \log_2 7 = 2.807$$

$$\therefore T(n) = n^{2.807}$$

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$$\in \Theta\left(n^{\log_2 7}\right) = \Theta\left(n^{2.807}\right)$$

Strassen's matrix multiplication time Complexity.